

STRUCTURA SI ORGANIZAREA CALCULATOARELOR

Calculatoare 3 2025-2026

CURS 7 Sapt 7 11 noiembrie 2025 14:00-16:00

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Modul de notare:

Examen:	50%	
Proiect	20%	A3
Laborator:	20%	A2
Test (Midterm):	10%	A1

Obs. Fiecare componenta a notei finale (A1, A2, A3 si Examen) trebuie sa fie promovata cu nota minima 5.

Planificarea activitatilor:

Laborator:

- saptamanal, conform orarului;
- sapt. 13, 14 refaceri laboratoare;

Proiect:

- conform orarului

Curs:

- saptamanal, conform orarului;

Test (Midterm): in saptamana a 10-a (9 decembrie 2025)

Continut

Bazele aritmetice ale calculatoarelor electronice

- Sisteme de numeratie

- Reprezentarea numerelor in virgula fixa / mobila

- Operatii aritmetice in virgula fixa / mobila

Proiectarea blocurilor logice pentru operatiile aritmetice in calculatoarele electronice

- Adunarea; scaderea

- Inmultirea

- Impartirea

Proiectarea procesoarelor

- Procesoare RISC

- Setul de instructiuni

- Calea de date (Data Path)

- Calea de control (Control Path)

Bibliografie

David A. Patterson, John L. Hennessy - Computer Organization and Design The Hardware Software Interface [RISC-V Edition] (6th edition Morgan Kaufmann, 2018)

David Harris, Sarah Harris - Digital Design and Computer Architecture (2007)

J. Hennessy, D. Patterson - Computer Architecture. A Quantitative Approach (6th edition Morgan Kaufmann 2019)

William Stallings - Computer Organization and Architecture Designing for Performance (10th edition Pearson 2016)

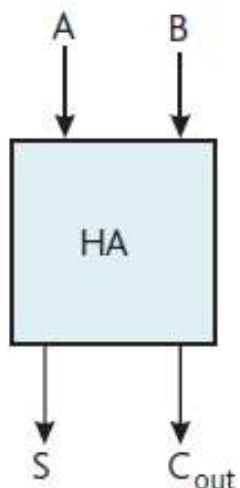
A. Tanenbaum, T. Austin - Structured Computer Organization, (6th edition Pearson 2013)

Karbo M.B. - A complete illustrated guide to the PC hardware (2005)

CIRCUITE PENTRU EFECTUAREA OPERATIILOR ARITMETICE IN PROCESOARE

Circuite pentru efectuarea operatiei de adunare

Semisumatorul – Half Adder



$$C_{OUT} = A \cdot B = \overline{\overline{A \cdot B}} = \overline{\overline{A} \cdot \overline{B}} = \overline{A + B}$$

$$S = \overline{A \cdot B} + A \cdot B = A \oplus B$$

$$S = \overline{\overline{A \cdot B} + A \cdot B} = \overline{(\overline{A \cdot B}) \cdot (A \cdot B)}$$

$$S = \overline{(\overline{A \cdot B}) \cdot (A \cdot B)} = \overline{\overline{A + B}} + \overline{\overline{A} + B} = A + B$$

Circuit Logic Combinational (CLC)

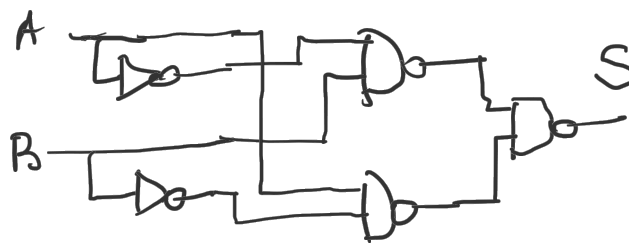
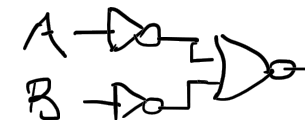
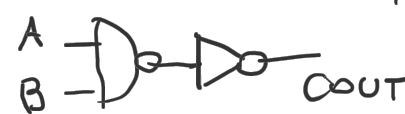
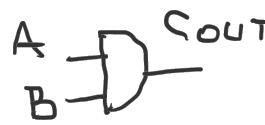
A	B	Cout	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

A, B - operanzi pe 1 bit

Cout - transport (carry)

S - suma

Cout, S - 1 bit

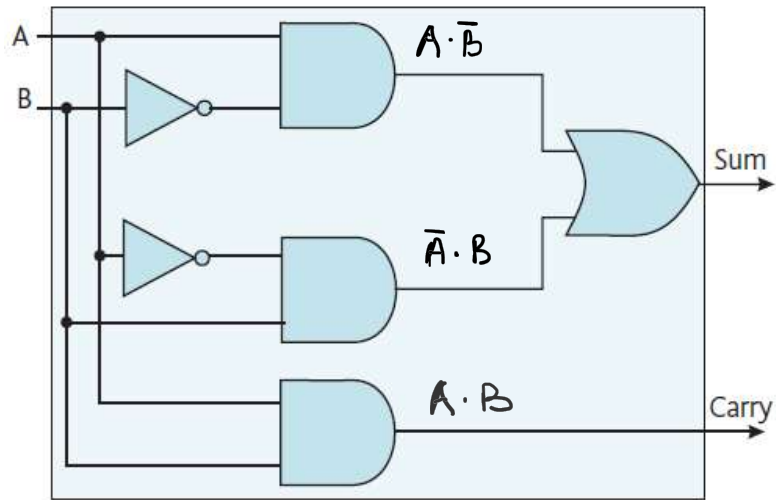


De Morgan

$$\begin{cases} \overline{x \cdot y} = \overline{x} + \overline{y} \\ \overline{x + y} = \overline{x} \cdot \overline{y} \end{cases}$$

$$\begin{cases} \cdot \rightarrow \text{SI} \\ + \rightarrow \text{SAU} \end{cases}$$

Solutii de implementare Semisumator

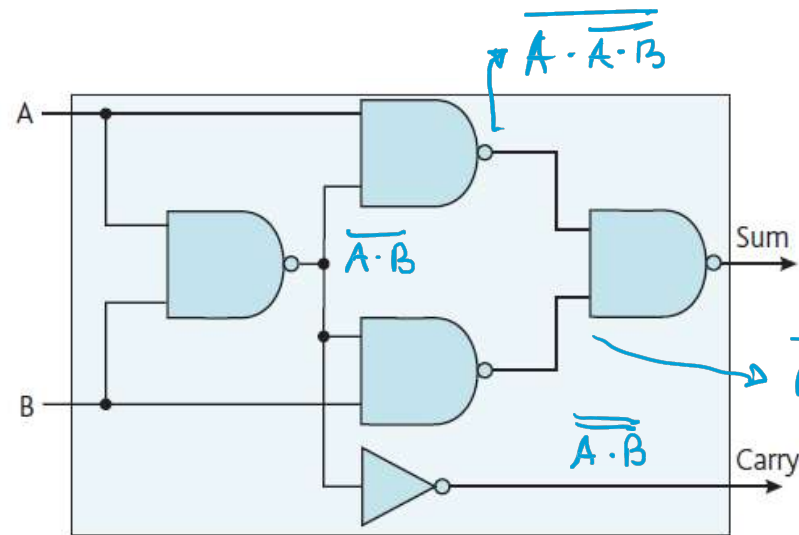
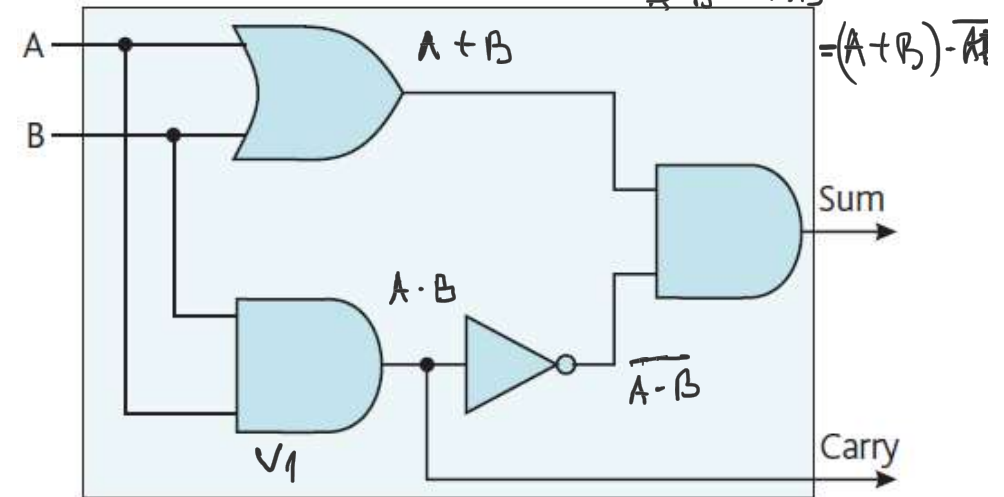


v1.

$$S = (A+B) \cdot (\overline{A \cdot B}) = \overline{A+B} + (\overline{A+B})^c$$

$$= \overline{A} \cdot \overline{B} + (A \cdot B) =$$

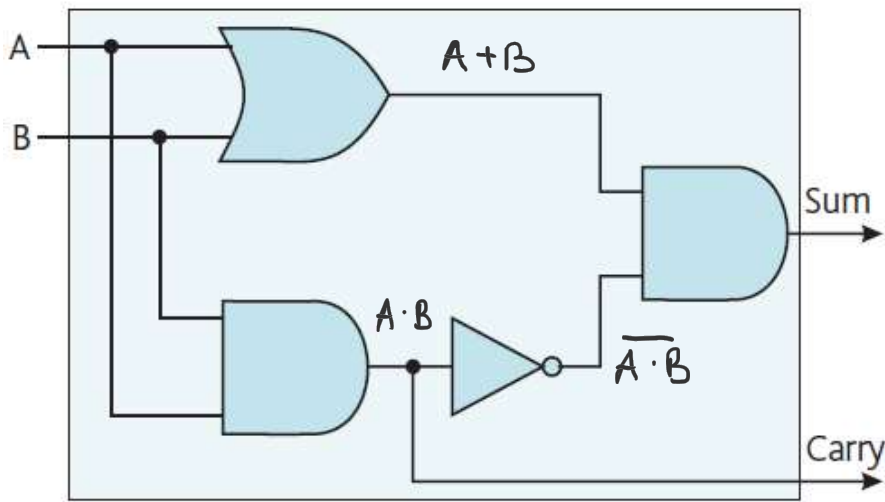
$$= \overline{A \cdot B} \cdot \overline{A \cdot B} =$$



V3

$$\overline{A \cdot \overline{AB}} = \overline{A} + AB$$

$$= \bar{A} \cdot B + A \bar{B} = A \oplus B$$



$$S = (A+B) \cdot \overline{A \cdot B} = \overline{\overline{(A+B)} \cdot \overline{A \cdot B}} = \overline{\overline{A+B} + A \cdot B} =$$

$\uparrow$ double négative $\uparrow$ de Morgan
 $(\overline{x \cdot y} = \overline{x} + \overline{y})$

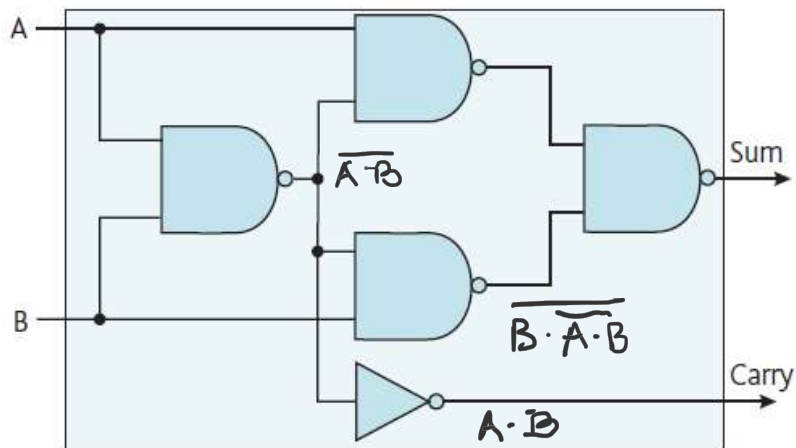
$$= \overline{\overline{A \cdot B} + A \cdot B} = \overline{\overline{A \cdot B}} \cdot \overline{A \cdot B} =$$

$\uparrow$ de Morgan $\uparrow$ de Morgan

$$= (A+B) \cdot \overline{A \cdot B} = (A+B) \cdot (\overline{A} + \overline{B}) =$$

$\uparrow$ de Morgan $\uparrow$ de Morgan

$$= \underbrace{A \cdot \overline{A}}_0 + A \cdot \overline{B} + \overline{A} \cdot B + \underbrace{B \cdot \overline{B}}_0 = A \cdot \overline{B} + \overline{A} \cdot B = A \oplus B$$



$$S = \overline{\overline{A \cdot \overline{A \cdot B}} \cdot \overline{B \cdot \overline{A \cdot B}}} = \overline{\overline{A + AB} \cdot \overline{B + AB}} =$$

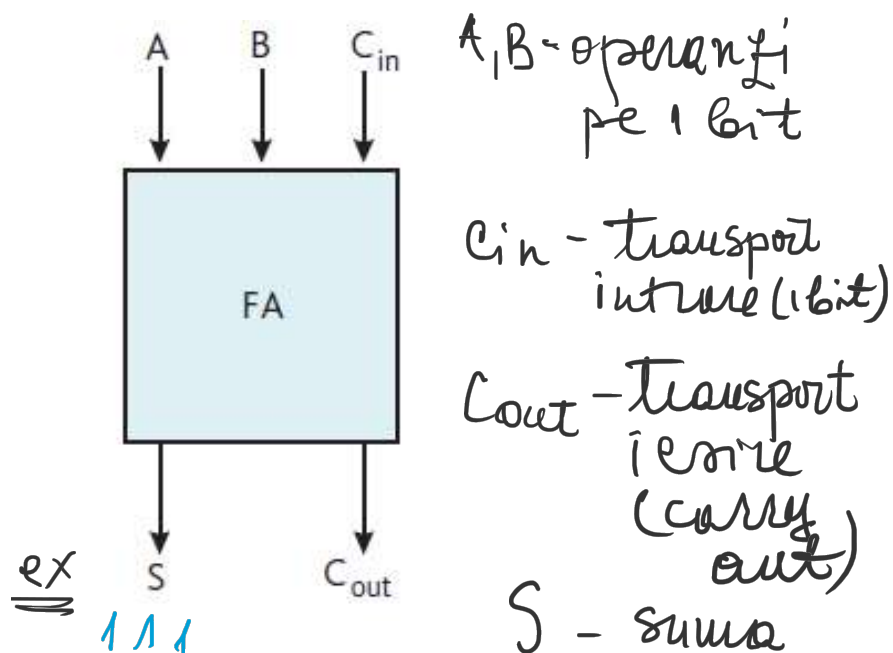
$\uparrow$ DM $\uparrow$ DM $\uparrow$ DM

$$= \overline{\overline{A} + AB + \overline{B} + AB} = \overline{A \cdot \overline{AB} + B \cdot \overline{AB}} =$$

$$= \overline{A \cdot (\overline{A} + \overline{B}) + B(\overline{A} + \overline{B})} = \overline{AA + AB + \overline{A}B + BB} =$$

$$= \overline{AB + \overline{A}B} = A \oplus B$$

Sumatorul complet elementar- Full Adder



A	B	C _{in}	S (sum)	C _{out} (carry-out)
0	0	0	0	0
0	1	0	1 *	0
1	0	0	1 *	0
1	1	0	0	1 *
0	0	1	1 *	0
0	1	1	0	1 *
1	0	1	0	1 *
1	1	1	1 *	1 *

ex

```

  111
1011 *
1101
-----
11000
  
```

Half Adder

Full Adder

adunare

$$\begin{cases}
 S = \bar{A} \cdot \bar{B} \cdot \bar{C}_{in} + \bar{A} \cdot B \cdot \bar{C}_{in} + A \cdot \bar{B} \cdot \bar{C}_{in} + A \cdot B \cdot \bar{C}_{in} \\
 C_{out} = A \cdot B \cdot \bar{C}_{in} + \bar{A} \cdot B \cdot C_{in} + A \cdot \bar{B} \cdot C_{in} + A \cdot B \cdot C_{in}
 \end{cases}$$

$X = \underbrace{X}_{SAU} + \underbrace{X}_{SAU} = \underbrace{X}_{SAU} + \underbrace{X}_{SAU} + \underbrace{X}_{SAU}$

$$C_{out} = \underbrace{A \cdot B \cdot \bar{C}_{in} + A \cdot B \cdot C_{in}}_{A \cdot B} + \underbrace{\bar{A} \cdot B \cdot C_{in} + A \cdot \bar{B} \cdot C_{in}}_{B \cdot C_{in}} + \underbrace{A \cdot \bar{B} \cdot \bar{C}_{in} + A \cdot B \cdot \bar{C}_{in}}_{A \cdot (\bar{C}_{in} + C_{in})}$$

$$C_{out} = A \cdot B \cdot (\bar{C}_{in} + C_{in}) + B \cdot C_{in} (\bar{A} + A) + A \cdot C_{in} (\bar{B} + B) \quad (y + \bar{y} = 1)$$

$$C_{out} = A \cdot B + B \cdot C_{in} + A \cdot C_{in}$$

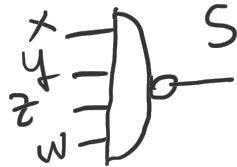
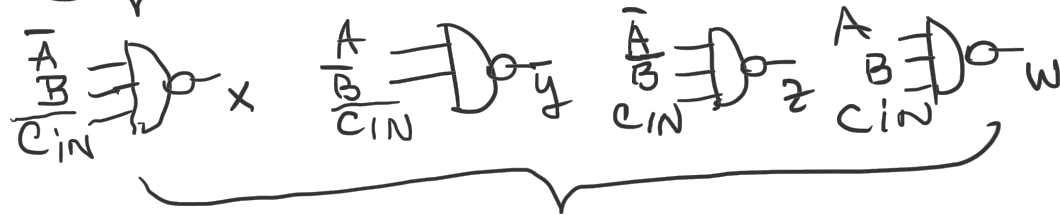
$$C_{OUT} = A \cdot B + B \cdot C_{IN} + A \cdot C_{IN}$$

$$S = \overline{A} \cdot B \cdot \overline{C_{IN}} + A \cdot B \cdot \overline{C_{IN}} + \overline{A} \cdot B \cdot C_{IN} + A \cdot B \cdot C_{IN}$$

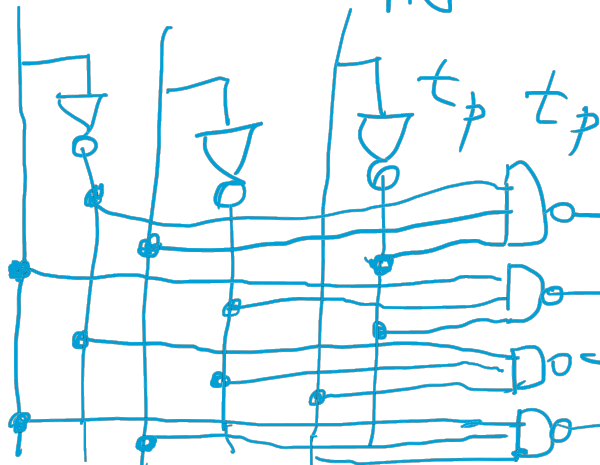
$$C_{OUT} = \overline{\overline{A \cdot B + B \cdot C_{IN} + A \cdot C_{IN}}} \quad \underline{\underline{DM}} \quad \overline{\overline{A \cdot B} \cdot \overline{B \cdot C_{IN}} \cdot \overline{A \cdot C_{IN}}}$$

$$S = \overline{\overline{A \cdot B \cdot \overline{C_{IN}}} + \overline{A \cdot B \cdot C_{IN}} + \overline{\overline{A} \cdot B \cdot C_{IN}} + \overline{A \cdot B \cdot C_{IN}}} \quad \underline{\underline{DM}}$$

$$= \overline{\overline{A \cdot B \cdot \overline{C_{IN}}} \cdot \overline{A \cdot B \cdot C_{IN}} \cdot \overline{\overline{A} \cdot B \cdot C_{IN}} \cdot \overline{A \cdot B \cdot C_{IN}}}$$

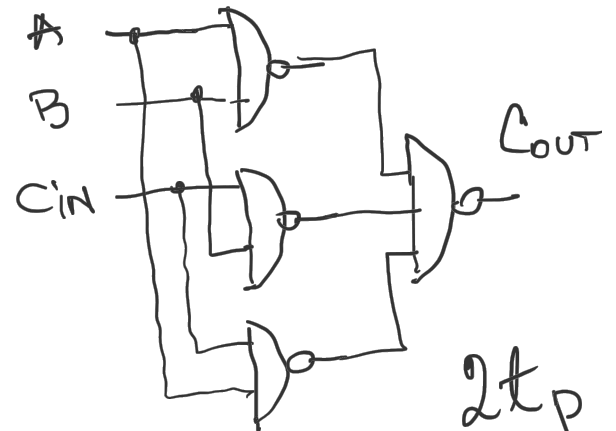


A B C_{IN}

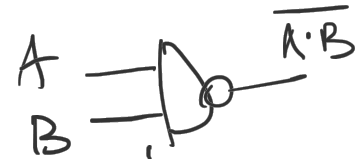


$$t_p \Rightarrow 3t_p$$

8 ports



2t_p



4 ports



$$t_p = t_1 - t_0$$

(time propagation)

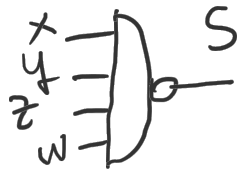
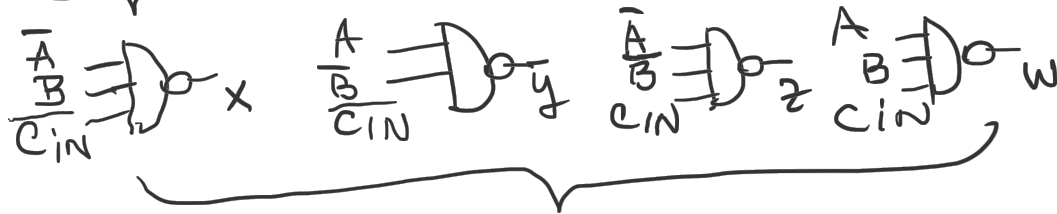
DM - relative

de Morgan

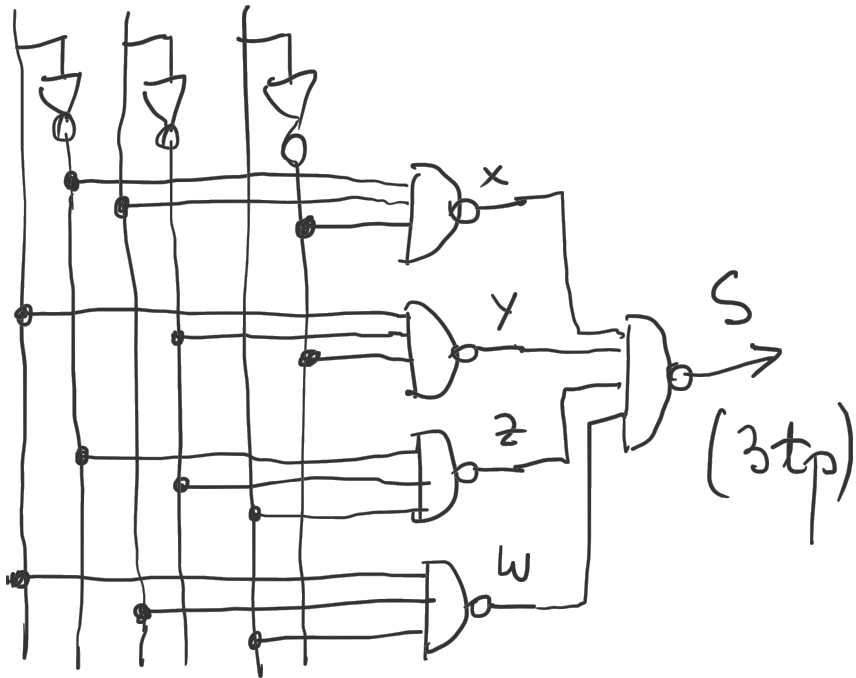
$$\overline{X + Y} = \overline{X} \cdot \overline{Y}$$

$$S = \overline{A} \cdot B \cdot \overline{C_{IN}} + A \cdot \overline{B} \cdot \overline{C_{IN}} + \overline{A} \cdot \overline{B} \cdot C_{IN} + A \cdot B \cdot C_{IN} = \text{DM}$$

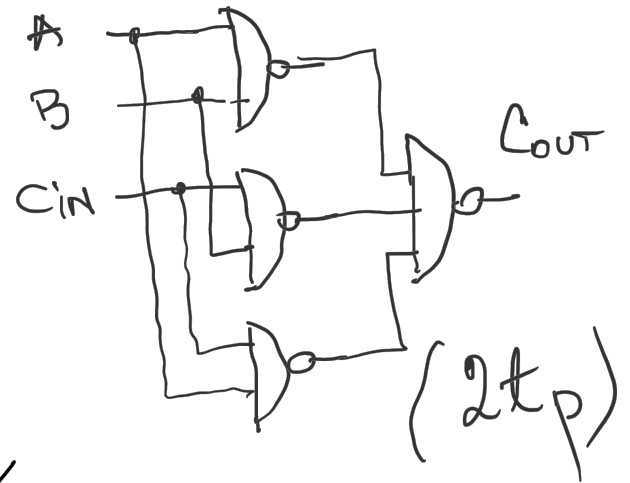
$$= \overline{A} \cdot B \cdot \overline{C_{IN}} \cdot A \cdot \overline{B} \cdot \overline{C_{IN}} \cdot \overline{A} \cdot \overline{B} \cdot C_{IN} \cdot A \cdot B \cdot C_{IN}$$



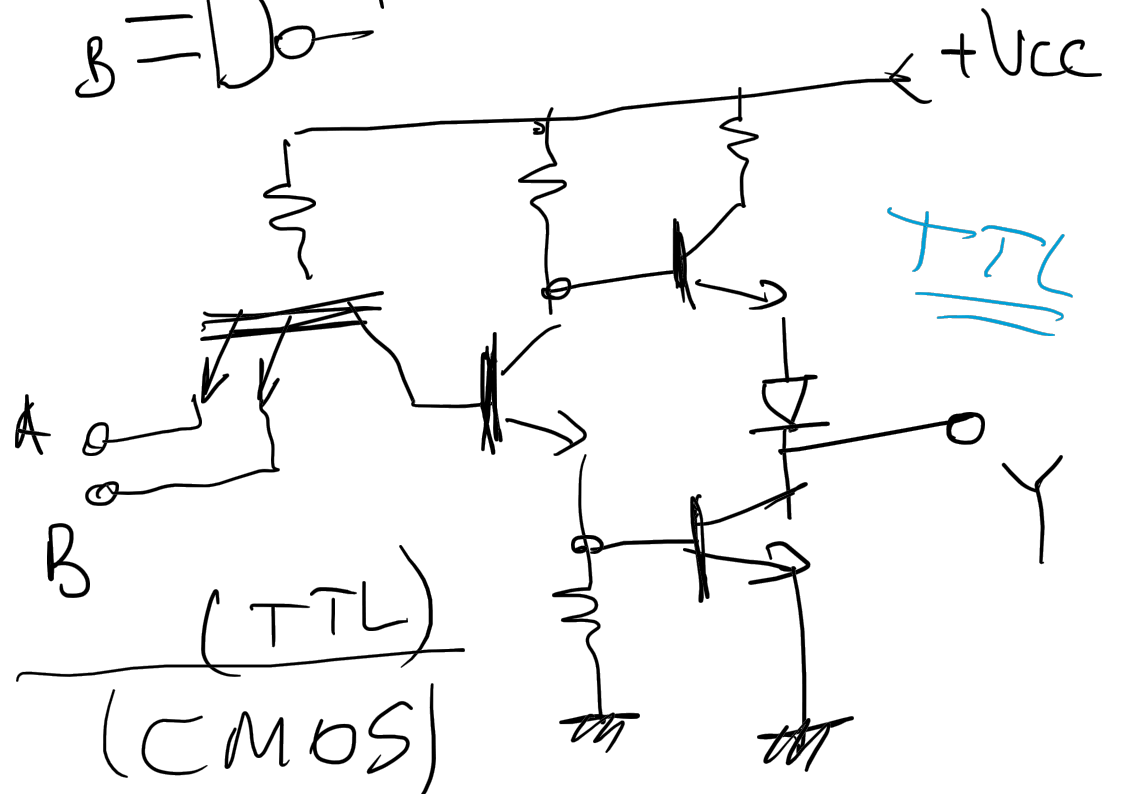
A B C_{IN}



$$C_{OUT} = \overline{A} \cdot B \cdot \overline{B} \cdot C_{IN} \cdot \overline{A} \cdot C_{IN}$$

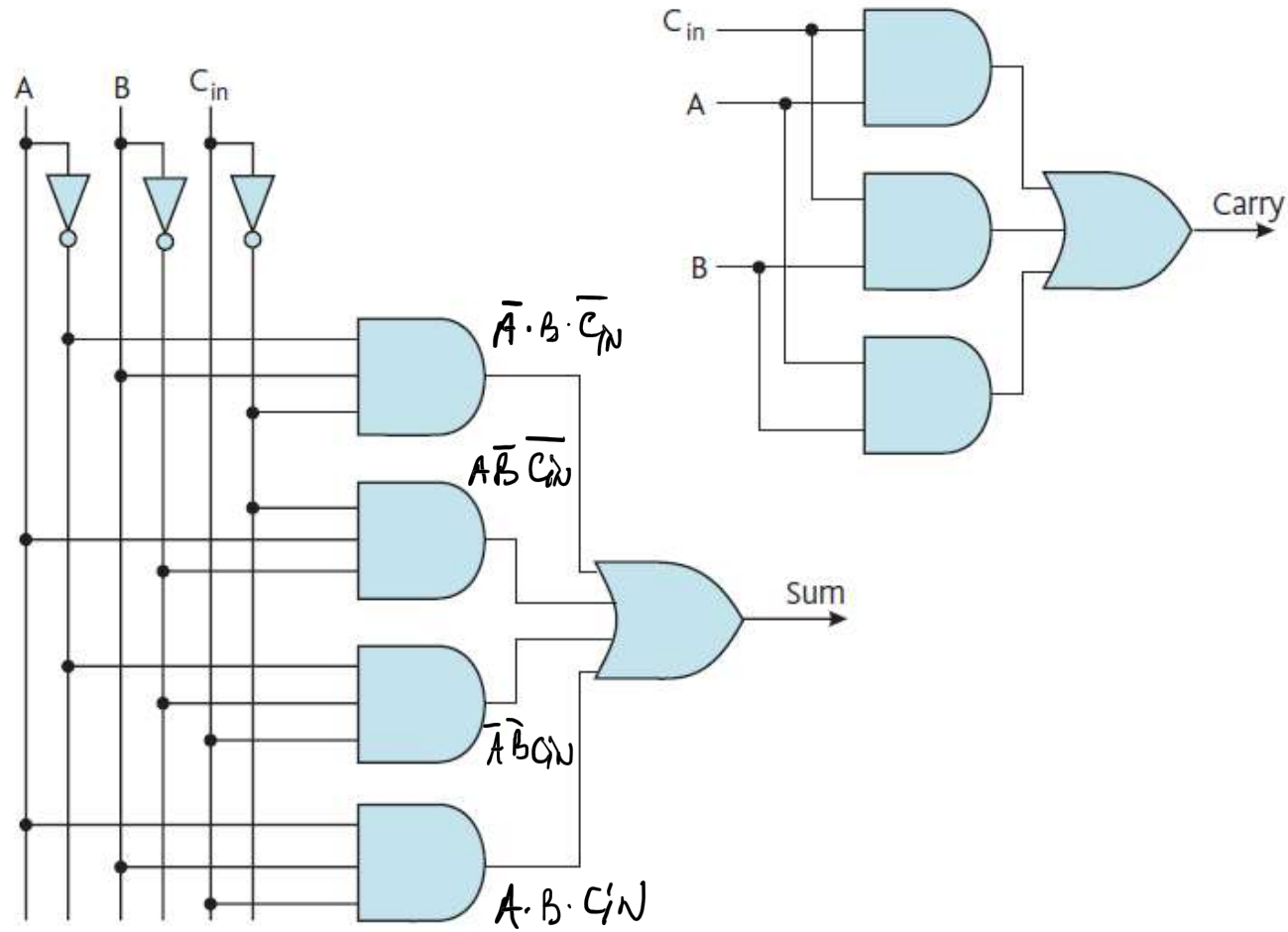


$$A \cdot B = Y$$

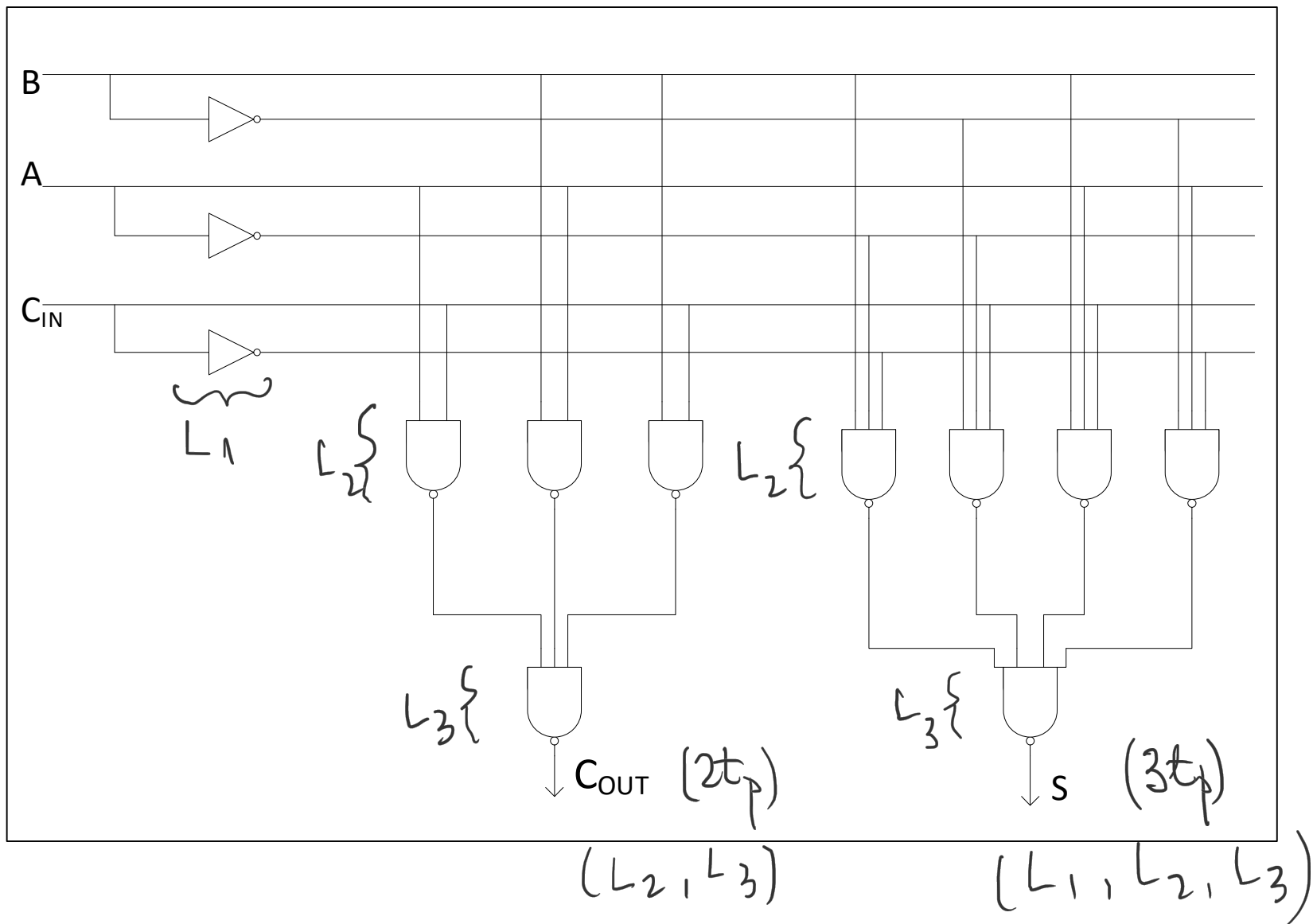


(CMOS)

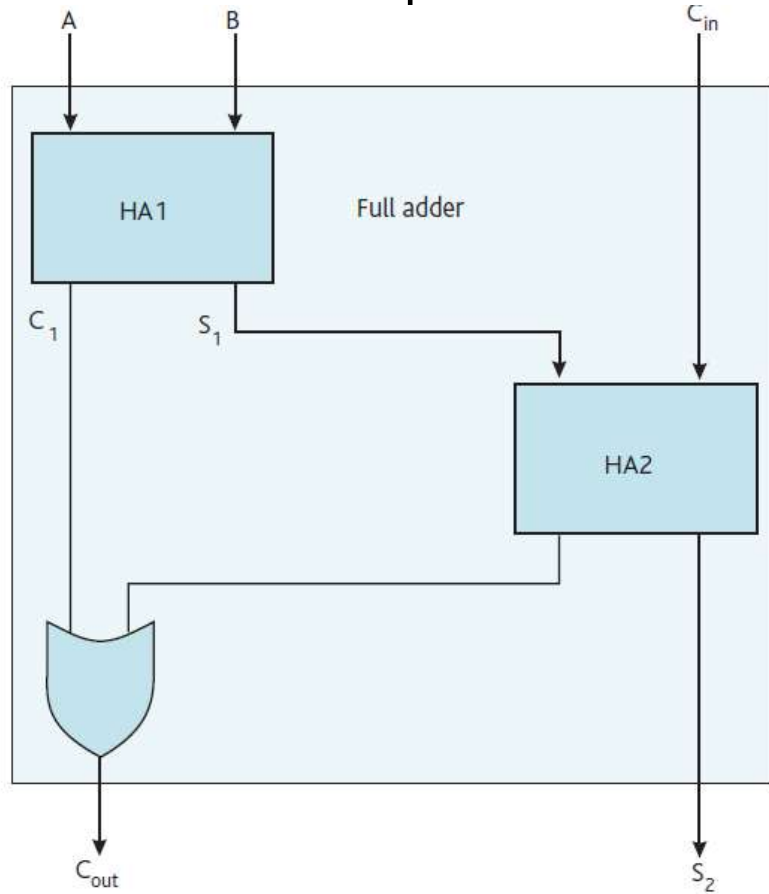
Solutie de implementare Sumator Complet



Solutie de implementare Sumator Complet



Solutie de implementare Sumator Complet folosind Semisumator

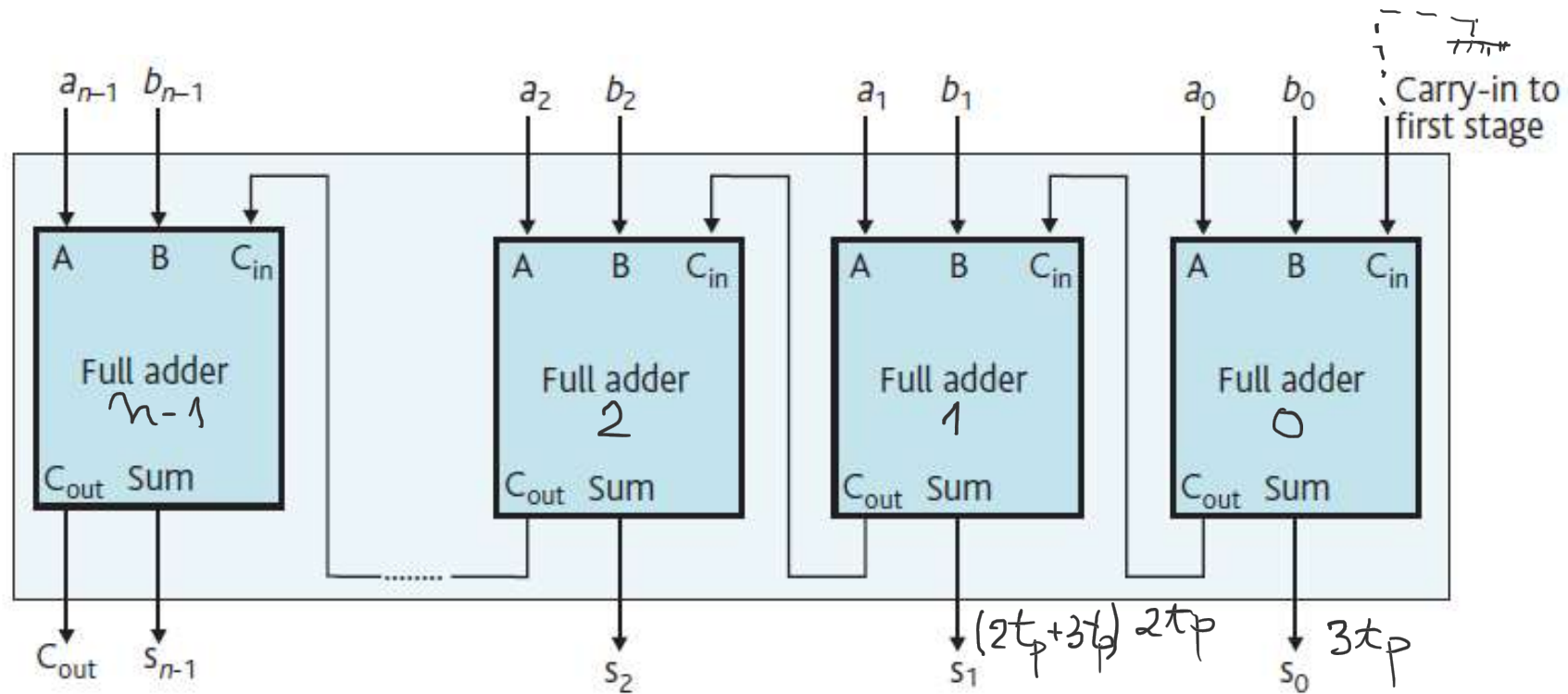


C_{in}	A	B	S_1	C_1	S_2	C_2	C_{out}
0	0	0	0	0	0	0	0
0	0	1	1	0	1	0	0
0	1	0	1	0	1	0	0
0	1	1	0	1	0	0	1
1	0	0	0	0	1	0	0
1	0	1	1	0	0	1	1
1	1	0	1	0	0	1	1
1	1	1	0	1	1	0	1

Handwritten annotations below the table:

- Under the first three columns (A, B, S_1), a bracket is drawn with the handwritten text "Semisumator 1".
- Under the next three columns (C_1 , S_2 , C_2), a bracket is drawn with the handwritten text "Semisumator 2".
- Below "Semisumator 1", the number "1" is written, with an arrow pointing to a circled S_1 .
- Below "Semisumator 2", the number "2" is written, with an arrow pointing to a circled C_{out} .

Sumator Paralel – Ripple-Carry Adder (sumator cu transport propagat)



$A(a_{n-1} a_{n-2} \dots a_2 a_1 a_0)$ – n biti

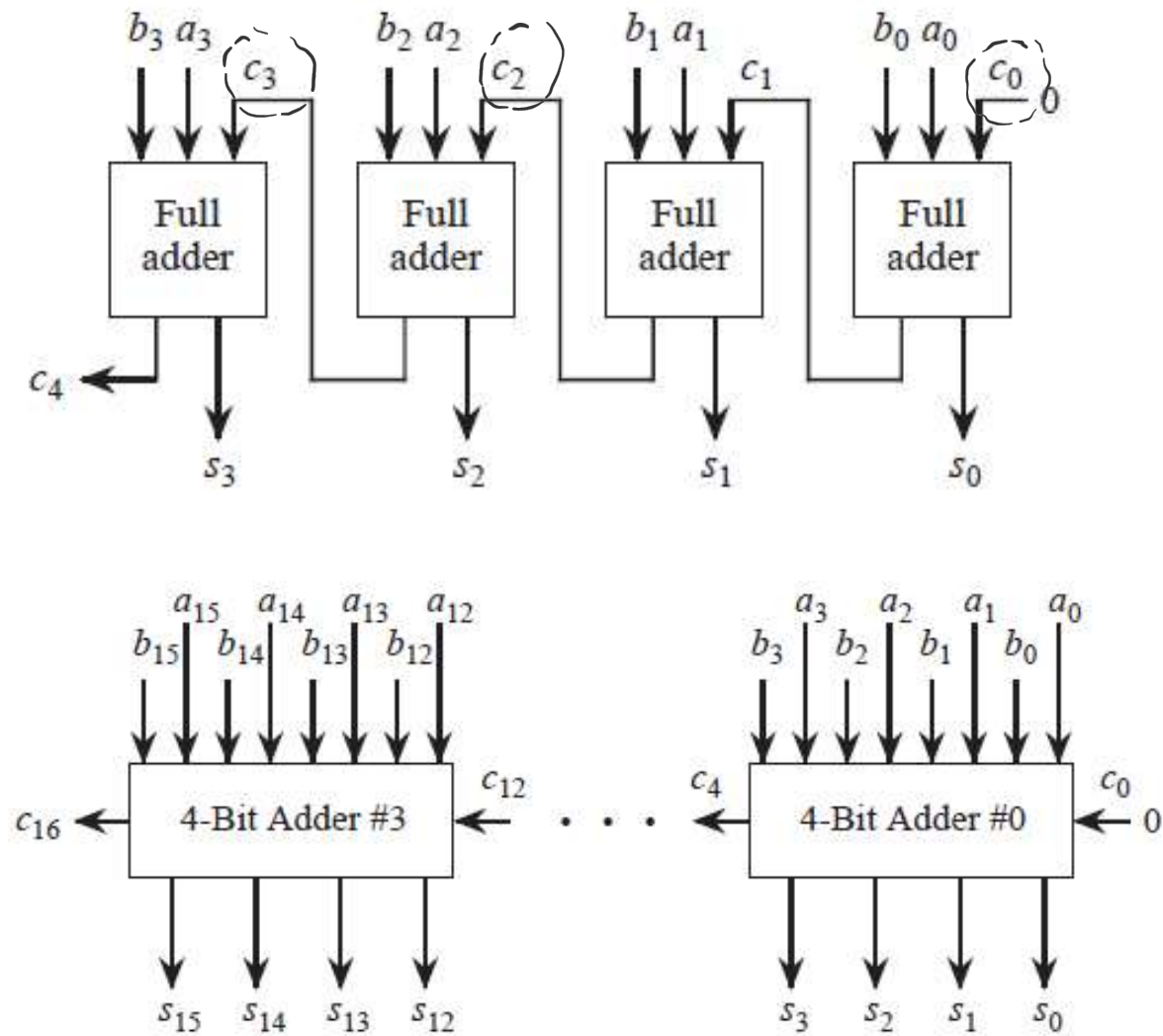
$B(b_{n-1} b_{n-2} \dots b_2 b_1 b_0)$ – n biti

$$\begin{cases} S_1 \rightarrow 2t_p + 3t_p = 5t_p \\ C_1 \rightarrow 2t_p + 2t_p = 4t_p \\ S_2 \rightarrow 4t_p + 3t_p = 7t_p \\ C_2 \rightarrow 4t_p + 2t_p = 6t_p \\ S_3 \rightarrow 6t_p + 3t_p = 9t_p \\ C_3 \rightarrow 6t_p + 2t_p = 8t_p \end{cases}$$

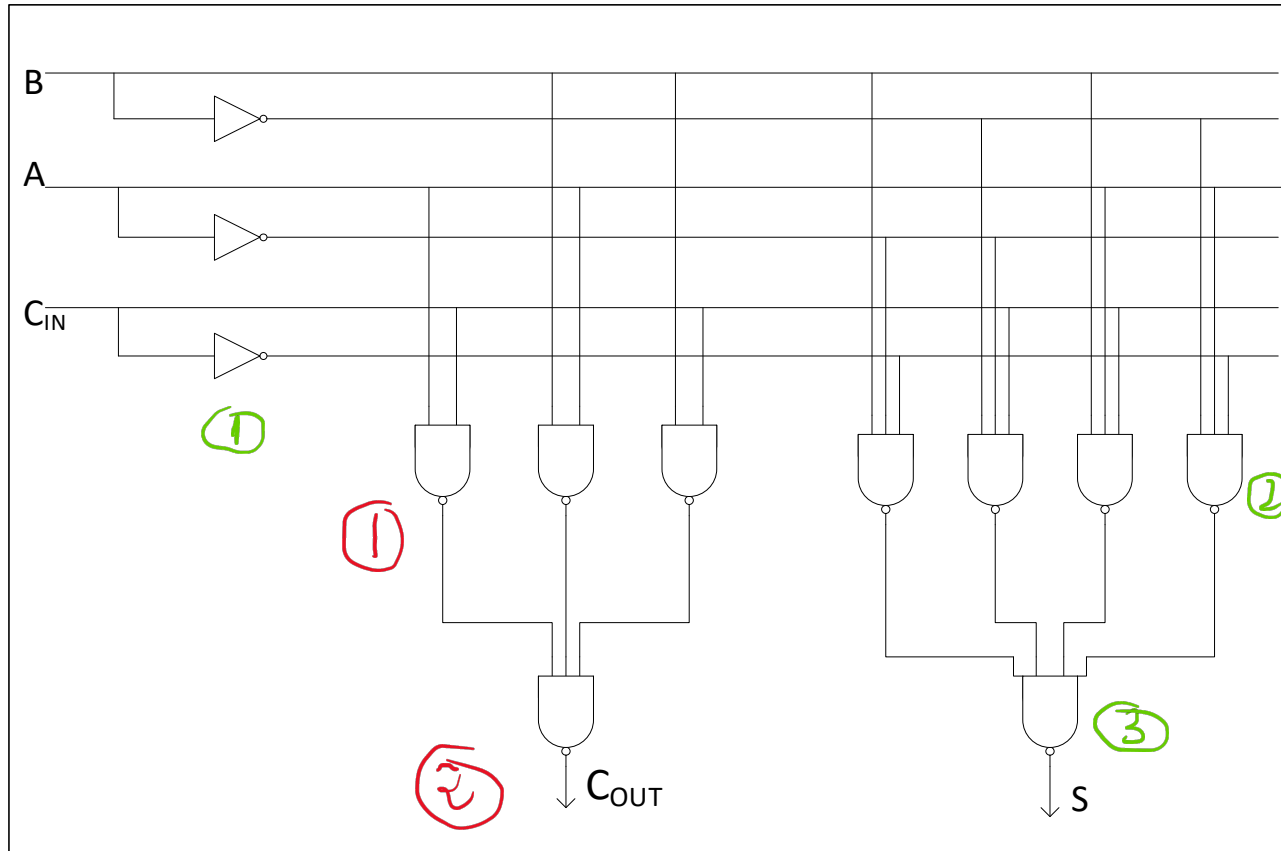
$$\begin{cases} C_n \rightarrow 2(n+1) \cdot t_p \\ S_n \rightarrow (2n+3) \cdot t_p \end{cases}$$

Portii logice $\rightarrow 12 \cdot n$

Sumator Paralel cu Transport Propagat – Ripple-Carry Adder



Solutie de implementare Sumator Complet



$$C \rightarrow 2t_p$$

$$S \rightarrow 3t_p$$

$$c_{i+1} = b_i c_i + a_i c_i + a_i b_i$$

$$= \overline{b_i c_i} \cdot \overline{a_i c_i} \cdot \overline{a_i b_i}$$

↑ De Morgan

(C_{OUT}) $c_{i+1} = b_i c_i + a_i c_i + a_i b_i$

$$c_{i+1} = a_i b_i + (a_i + b_i) c_i$$

$$G_i = a_i b_i \quad P_i = a_i + b_i$$

$$c_{i+1} = G_i + P_i c_i$$

$$c_{i+1} = (b_i \wedge c_i) \vee (a_i \wedge c_i) \vee (a_i \wedge b_i)$$

$$s_i = \overline{a_i} \overline{b_i} c_i + \overline{a_i} b_i \overline{c_i} + a_i \overline{b_i} \overline{c_i} + a_i b_i c_i$$

G - Generate

P - propagate

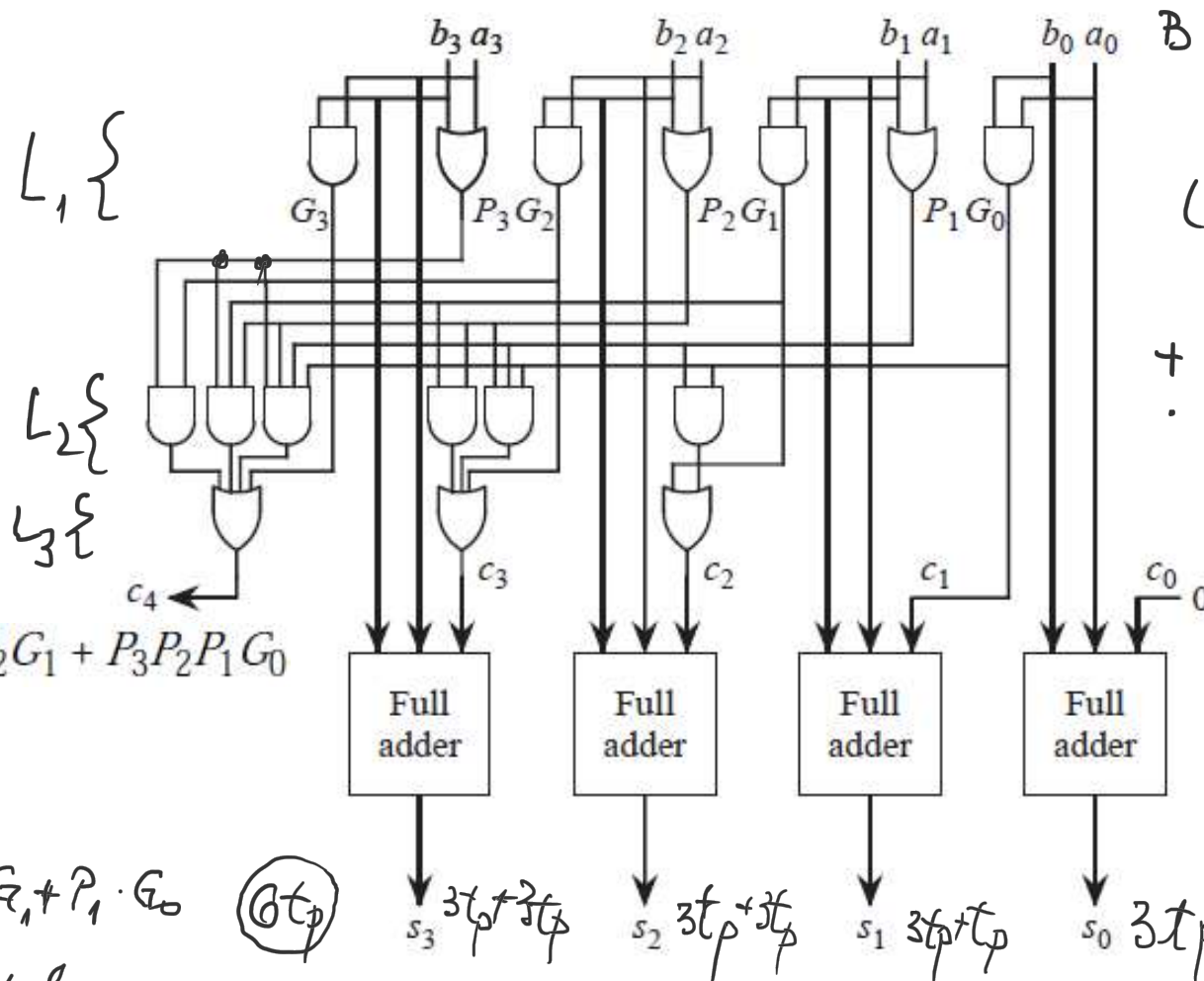
• - Si logic (1)

+ - SAU logic (V)

Sumator cu Transport Anticipat – Carry Look-Ahead Adder

$$\textcircled{*} \quad c_{i+1} = G_i + P_i c_i$$

**CARRY
LOOK
AHEAD
ADDER**



$A (a_3 a_2 a_1 a_0)$
 $B (b_3 b_2 b_1 b_0)$
 $A + B$
 (suma
 aritmetică)
 + SAU Logic
 · Si Logic

**END 11
 nov
 2025**

$$c_4 = G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 G_0$$

$$c_0 = 0$$

$$c_1 = G_0 = a_0 \cdot b_0$$

$$\textcircled{*} c_2 = G_1 + P_1 \cdot c_1 = G_1 + P_1 \cdot G_0 \quad (6t_p)$$

$$G_1 = a_1 \cdot b_1 \quad P_1 = a_1 + b_1$$

$$\textcircled{*} c_3 = G_2 + P_2 \cdot c_2 = G_2 + P_2 \cdot (G_1 + P_1 \cdot G_0) = G_2 + P_2 \cdot G_1 + P_2 \cdot P_1 \cdot G_0$$

$$c_4 = G_3 + P_3 \cdot c_3 = G_3 + P_3 \cdot (G_2 + P_2 \cdot G_1 + P_2 \cdot P_1 \cdot G_0) = G_3 + P_3 \cdot G_2 + P_3 \cdot P_2 \cdot G_1 + P_3 \cdot P_2 \cdot P_1 \cdot G_0$$

$$G_2 = a_2 \cdot b_2 \quad P_2 = a_2 + b_2$$

$$G_3 = a_3 \cdot b_3 \quad P_3 = a_3 + b_3$$